Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Thompson Jr, James N. Ph.D.		
Project Title:	Effect of hypergravity, vibration, and radiation on genetic and developmental stability in Drosophila melanogaster		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jthompson@ou.edu	**Fax:**	FY 405-325-6202
PI Organization Type:	UNIVERSITY	**Phone:**	405-325-4821
Organization Name:	University of Oklahoma		
PI Address 1:	Department of Zoology		
PI Address 2:	RH 314		
PI Web Page:	http://faculty-staff.ou.edu/		
City:	Norman	**State:**	OK
Zip Code:	73019-3032	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	08/01/2000	**End Date:**	01/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	10	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1427		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This project is designed to determine the effects of three potential stresses associated with long-term activity on the International Space Station (ISS): hypergravity, vibration, radiation, and some of their possible interactions. Experiments are conducted with Drosophila melanogaster as the model organism. There are three primary goals. The first is to measure the effect of hypergravity, vibration, and radiation using large-scale assays of mutation rate for sex-linked lethal loci, somatic mutation, aneuploidy, chromosome breakage, and other genetic changes. These genetic changes are caused by somewhat different mechanisms and together can give a clearer understanding of the relative sensitivities of these processes to potentially harmful stress conditions. The breeding programs have been planned to allow identification of clusters of identical new mutation due to premeiotic events, so we are able to distinguish between premeiotic and meiotic effects on genetic stability. The second goal is to evaluate the effect of these environmental conditions on developmental stability, by quantifying changes in fluctuating asymmetry, spatial patterning of peripheral sensory structures, and reductions in relative fitness, viability, and longevity. The third goal is to develop and test modified versions of these		

	experimental assays, such as mutation accumulation protocols, that can be used in long-term studies on the International Space Station and that can maximize the information on different elements of genetic and developmental stability that can be collected from the same treated individuals. These experiments, therefore, serve both as an in-depth analysis of genetic and developmental stability under these selected environmental conditions and as a feasibility study for testing for the same effects on the International Space Station where these factors can interact with each other and with microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will contribute to our understanding of mutation and developmental responses to stress. The results also form ground-based data sets that will serve as controls for interpreting results from mutation accumulation experiments in space.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 08/08/2019)